

Jonathan Cruz

SENIOR BACKEND & PAYMENTS ENGINEER

📍 SÃO PAULO, BRAZIL

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PROFILE

"Senior software engineer with over 10 years of experience building fintech and payments infrastructure. Deep expertise in transactional Postgres, double-entry ledgers, KYC/EDD compliance, and integrations with banking and identity providers (Plaid, Treasury Prime). Strong TypeScript across the stack with NestJS, Prisma, and event-driven architectures designed for fault tolerance, idempotency, and auditability."

PROFESSIONAL EXPERIENCE

SELECTED ROLES & CONTRIBUTIONS

FINTECH FOR VC FUND
ADMINISTRATION

Sydecar

FULL-STACK SOFTWARE
ENGINEER

Double-Entry Ledger

POSTGRES

Contributed to a proprietary double-entry ledger engine (Fragment) with append-only immutable entries, deterministic idempotency keys (), derived balances (never stored as source of truth), and minor-unit integer accounting with banker's rounding. Posting flow is event-driven and fully reconstructable from source events.

POSTGRES

LEDGER

DOUBLE-ENTRY

DECIMAL

Plaid + Treasury Prime

BANKING

Worked on Plaid Link integration (createLinkToken → exchangePublicToken → access_token) with processor tokens powering ACH transfers via Treasury Prime. Built a mock banking engine for the local sandbox with idempotency tracking by application/account IDs.

PLAID

TREASURY PRIME

ACH

BANKING

KYC/EDD

COMPLIANCE

Designed and implemented Enhanced Due Diligence including LP funding blocks, UBO verification, FinCEN integration, CTA compliance, and automatic KYC revalidation for active investors. Built race-resistant verify-identity flow with re-check inside a to handle "email-free-at-blur → taken-at-submit" edge cases (409 EMAIL_NOW_EXISTS).

COMPLIANCE

KYC

UBO

FINCEN

Webhook Idempotency

CONCURRENCY

Implemented webhook dedup via (unique) with PROCESSING/PROCESSED/REJECTED state machine, and an API-level idempotency interceptor with 48h TTL scoped by .

IDEMPOTENCY

WEBHOOKS

CONCURRENCY

Fault Tolerance

BULLMQ OUTBOX

Standardized event delivery using BullMQ Pro with exponential backoff (5 retries / 3s base in ledger; 20 retries / 10s base in pubsub). Outbox pattern: persist resolution first, enqueue after — jobs are idempotent and survive enqueue failure. Global per-deal ordering via Redis INCR counters with deterministic predecessor lookup ().

BULLMQ

OUTBOX

REDIS

FAULT-TOLERANCE

Billing V2

ARCHITECTURE

Designed (ADR) and implemented fund billing V2 with quarterly fee scheduler, billing model versioning, and administrative fee structure management.

TS-REST

ZOD

NESTJS

QuickBooks

INTEGRATION

Built bidirectional invoice sync engine with mock service for local dev, state validation, auto-sync on approval, and full E2E test suite with Playwright and Cucumber BDD.

QUICKBOOKS API

PLAYWRIGHT

CUCUMBER

ts-rest

TYPE-SAFE
CONTRACTS

Migrated legacy controllers (auth, onboarding, admin) to type-safe ts-rest contracts with Zod schemas, improving type safety end-to-end.

TYPESCRIPT

ZOD

TS-REST

MAY 2023 — PRESENT
SAN FRANCISCO, CA (REMOTE)

B2B PAYMENTS & WHOLESALE Zax SR SOFTWARE ENGINEER JAN 2019 — MAY 2023 SÃO PAULO	B2B Payments MOBILE Built from scratch a wholesale ordering app with integrated payment flows and credit-limit validation for business partners, using React Native, TypeScript, and GraphQL. REACT NATIVE TYPESCRIPT GRAPHQL Serverless AWS Deployed full cloud infrastructure on AWS (EC2, ECS, S3, Lambda, VPCs) with serverless web apps using React and Next.js. AWS NEXT.JS SERVERLESS CI/CD AUTOMATION Configured end-to-end delivery pipeline with GitHub Actions, Fastlane, and Code Climate for quality control. Docker for local dev. GITHUB ACTIONS FASTLANE DOCKER
CRYPTO CONSUMER FINTECH Netlolo SR SOFTWARE ENGINEER SEP 2018 — JAN 2019 SÃO PAULO	Crypto Wallet WEB3 Built a mobile + web app bringing cryptocurrency to mainstream users with React Native and TypeScript, integrating Web3 smart contracts on Ethereum for self-custodial wallets and on-chain transactions. REACT NATIVE TYPESCRIPT WEB3 ETHEREUM Chat SDK REAL-TIME Developed a chat SDK for businesses using React Native and React/Redux, powered by XMPP and a Java backend for real-time messaging. REACT REDUX XMPP JAVA Zero DOWNTIME Built CI/CD automations ensuring continuous deployment with no service interruption. PostgreSQL with Hasura as the GraphQL layer. POSTGRESQL HASURA CI/CD
CONSUMER HEALTHCARE Escale SR SOFTWARE ENGINEER JAN 2018 — SEP 2018 SÃO PAULO	Healthcare Platform REACT Built the product foundation with React, Redux, and a Node.js/TypeScript backend for healthcare services sold via Google Ads, including lead capture, eligibility checks, and provider matching. REACT REDUX NODE.JS TYPESCRIPT Mentorship TECH TALKS Led internal Tech Talks on RESTful design, Functional Programming, and Microservices architecture. MySQL, PostgreSQL, and MongoDB depending on service responsibility. MICROSERVICES CIRCLECI MONGODB
STUDENT LOANS FINTECH PraValer PL SOFTWARE ENGINEER JUN 2017 — JAN 2018 SÃO PAULO	Loan Origination NODE Migrated legacy PHP loan origination platform to a Node.js/Express backend for a student loans product with governmental funding, handling loan request lifecycle, eligibility, and disbursement flows. NODE.JS EXPRESS LOANS PHP Database OPTIMIZATION Restructured PostgreSQL transactional tables with improved schemas and indexes for loan ledger entries. MongoDB for loan requests and logs. Frontend with React and Redux. POSTGRESQL MONGODB REACT REDUX

CORE SKILLS

PAYMENTS & FINTECH

Plaid, Treasury Prime, ACH, KYC/EDD, FinCEN, Double-entry ledgers, Idempotency, Webhooks

BACKEND

TypeScript, NestJS, Node.js, Prisma, PostgreSQL, ts-rest, Zod, GraphQL

RELIABILITY

BullMQ, Outbox pattern, Event ordering, Redis, Datadog

FRONTEND

React, Next.js, React Native, TanStack Query

INFRASTRUCTURE

AWS, GCP, Docker, CI/CD, Nx Monorepo

LANGUAGES

PORTUGUESE

Native

ENGLISH

C1/C2

EDUCATION

Anhanguera

BACHELOR OF TECHNOLOGY (B.TECH.) IN INFORMATION TECHNOLOGY ·
JAN 2016 — JUL 2018